

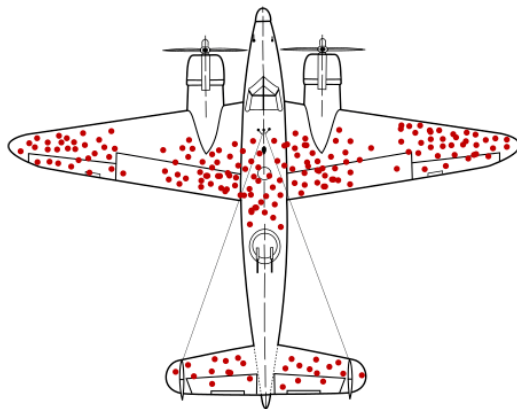
Causal Inference

Bias in Observational Data: Example

"During World War II, the statistician Abraham Wald took survivorship bias into his calculations when considering how to minimize bomber losses to enemy fire ..." (source [Wikipedia](#))

Below is an image showing where planes were shot. These data were collected once the plane returned to airbase.

Q: Where will you reinforce the armor?



Bias in Observational Data

1. **Survivor bias:** we only observe the cases that survive the selection process (what we've just seen)
2. **Self-reporting bias:** After major elections, surveys typically find that 60–70% of respondents say they voted, but actual turnout records show only ~50–55% turnout.
3. **Post-treatment bias:** Does a college education increase income?
Suppose we adjust for "occupation" when estimating the effect of college on income. But occupation is partly determined by having gone to college.
4. **Attrition bias:** Do online learning platforms improve performance?
If struggling students are more likely to drop out, the final test score average only reflects those who didn't drop out.

5. **Reverse causality:** Does stress cause lack of sleep?
Maybe. But also: lack of sleep causes stress. If we just look at correlation, we might mistake the direction of causality.
6. **Collider bias** (Berkson's Paradox): Is exercise unrelated to healthy eating?
Among hospital patients, those who exercise tend to eat less healthily — but this is not true in the general population.
Why? Hospital admission might depend on either bad diet or lack of exercise — conditioning on this collider creates a spurious relationship.

Simpson's Paradox: Compare Like With Like

Question. Which kidney-stone treatment is more successful?

Stone size	Treatment A	Treatment B	Within group
Small stones	81 / 87 = 93%	234 / 270 = 87%	A is higher
Large stones	192 / 263 = 73%	55 / 80 = 69%	A is higher

Treatment A looks better for both small stones and large stones.

Simpson's Paradox: The Trap

	Treatment A	Treatment B
All patients combined	273 / 350 = 78%	289 / 350 = 83%

Treatment A patients

87 small stones, 263 large stones
mostly harder cases

Treatment B patients

270 small stones, 80 large stones
mostly easier cases

Stone size is the hidden variable: it affects success, and the two treatments were used on different mixes of patients.

Overall rates can reverse the within-group story.

Spurious Correlations from Observational Data

Today we rely heavily on quantitative methods to recover (conditional) correlation or causal relationship from data.

But feeding data to untrained methods might yield [spurious correlation](#)...

Pursuit of Causation

However, it is often the causal inference that matters in the real world. For instance, we might be interested in the following questions.

- What would happen to the patient if they received treatment instead of ?
- What would happen to the unemployment rate if the U.S. government increased minimum wages?
- What would happen to the case number if a state took a different action in April?

In all these what-ifs, we notice that it is always comparing outcomes under different conditions for the same subject(s). In other words, causal inference is the comparison between potential outcomes under treatment and control for the same unit(s).

We will take a close look at the [potential outcome framework](#)

Applications of Potential Outcome

Using the potential outcome framework, we can describe the sources of bias and more:

- [Propensity score methods](#)
- [Instrumental variable analysis](#)